TECHNICAL REPORT / DATA RELEASE 0.2 / SEPTEMBER 2026

# Detector-domain characterization of thorium-argon calibration exposures

Recorded with EXOhSPEC using a ZWO CMOS-family detector

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| CONTEXT | EXOhSPEC instrumentation study |
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| EXPOSURES | 30, 60, 120, and 180 seconds |
| PUBLIC SCOPE | Detector coordinates; raw FITS and optical details withheld |

**Principal finding.** The 120 s exposure is the strongest single-frame compromise in this sequence. The full four-exposure set remains preferable for high-dynamic-range display and bright-core protection.

This report distinguishes measured detector morphology from atomic interpretation. No wavelength or species labels are assigned before a validated order trace and dispersion solution.

# ABSTRACT Summary

Thorium-argon (Th-Ar) hollow-cathode lamps are established wavelength references for astronomical spectrographs because they produce a dense forest of narrow, repeatable emission features. This report presents a detector-domain study of four Th-Ar exposures recorded with EXOhSPEC. The public analysis compares exposure times while withholding raw FITS files, complete headers, exact detector geometry, and laboratory optical details.

The unsaturated integrated response follows exposure time with $R^2 = 0.999276$ and a maximum fractional residual of 1.994%. Pixels at or above 60,000 ADU increase from 1 at 30 s to 121 at 180 s. Phase correlation gives a maximum translation estimate of 0.071 native pixel relative to 120 s under a translation-only model. The 120 s frame is therefore recommended when one exposure must represent the set, while an HDR estimator provides the clearest detector-space visualization.



Keywords: thorium-argon; hollow-cathode lamp; echelle spectroscopy; wavelength calibration; EXOhSPEC; CMOS; exposure optimization


| 120 s | 0.9993 | 1.994% | 0.071 px |
|---|---|---|---|
| recommended single frame | response $R^2$ | largest fit residual | largest estimated shift |

# 1 Purpose, scope, and privacy boundary

The project has two goals: explain what a Th-Ar source is and determine what the supplied EXOhSPEC exposure sequence supports scientifically before wavelength calibration. The central question is whether 120 s is sufficient and what is gained by retaining 30, 60, and 180 s frames.

This release identifies EXOhSPEC and a generic ZWO CMOS-family detector. It intentionally omits exact hardware configuration, optical prescriptions, laboratory layout, raw detector dimensions, full headers, file paths, and raw images. Public coordinates are normalized. This protects the laboratory while preserving the analysis logic and aggregate evidence.

# 2 Emission, absorption, and the hollow-cathode source

A Th-Ar lamp is a hollow-cathode discharge source. Argon ions bombard a thorium-bearing cathode and sputter thorium into the discharge. Collisions excite neutral and ionized species. Radiative relaxation produces photons at discrete energies, creating narrow emission features.

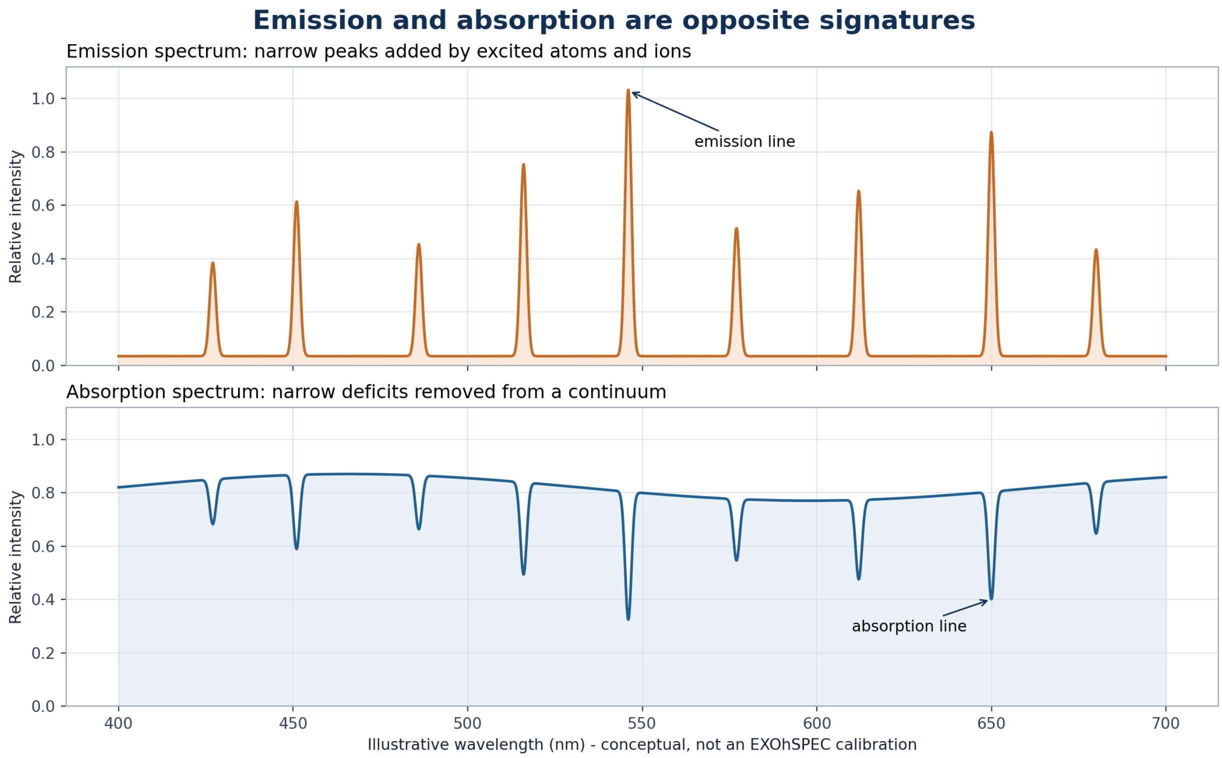


Figure 1. Conceptual distinction between emission peaks and absorption deficits. The wavelength values are illustrative and are not measured EXOhSPEC line assignments.

An emission line is a positive peak because the source adds photons at a transition wavelength. An absorption line is a deficit below a continuum because intervening matter removes photons. The EXOhSPEC lamp frame contains emission-line images; a stellar spectrum commonly contains absorption structure.

# 3 Why thorium and argon are used

Thorium supplies a dense pattern of narrow transitions. NIST Standard Reference Database 161 contains reference wavelengths for more than 20,000 Th I, Th II, and Th III features together with argon line lists. The dominant terrestrial isotope is thorium-232, whose approximately 1.40 x $10^{10}$ year half-life and dominance reduce isotopic complexity.

Argon sustains the discharge and adds Ar I, Ar II, and Ar III features. Some carrier-gas lines are very strong. This is useful for acquisition but challenging for detector dynamic range: faint thorium structure may require an exposure that pushes bright cores toward the ceiling.

USGS estimates about 10.5 mg $kg^{-1}$ thorium in upper continental crust. NOAA gives argon as about 0.934% by volume of dry air, dominated by argon-40. Abundance does not determine calibration quality; reference accuracy, line density, usable intensity, and stability do.

# 4 Reading the measured EXOhSPEC detector frame

The image below is derived from the measured exposure set. An inverse-hyperbolic-sine display stretch makes weak structure visible while retaining bright features. Quantitative calculations use linear detector values; the stretch affects appearance only.

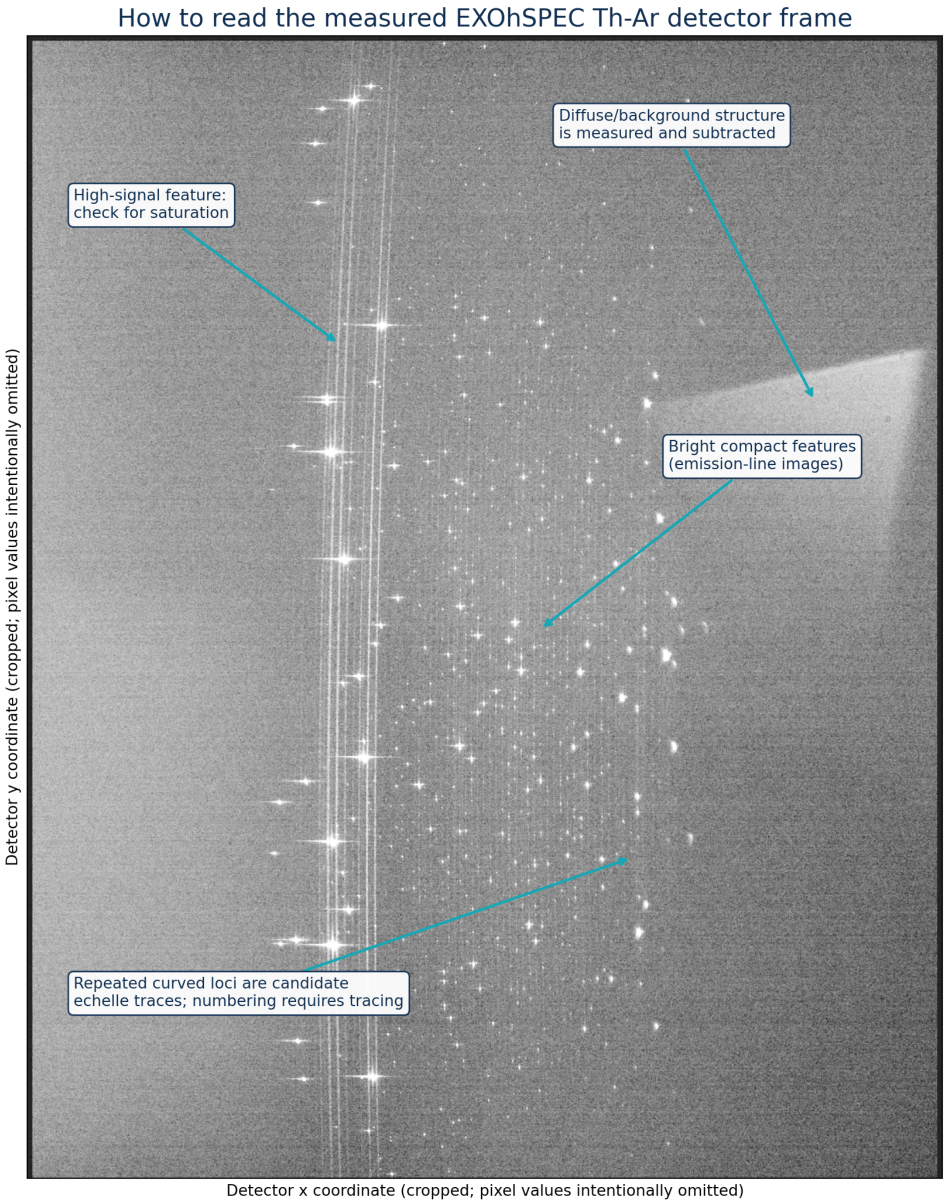


Figure 2. Measured detector morphology with a reading guide. Bright compact features are emission-line images. Repeated loci are candidate echelle traces, but order numbering and wavelengths require calibration.

# 5 Detector-domain methodology

## 5.1 FITS ingestion

The primary image is memory mapped and FITS scaling keywords are applied so calculations use physical unsigned detector values. Only exposure time is admitted to public metadata.

## 5.2 Robust background

The pedestal is the median of a sparse full-frame sample. Scatter is 1.4826 times the median absolute deviation, which is resistant to sparse bright features.

## 5.3 Public crop

Bright-pixel coordinate percentiles define a padded feature-rich region. Published axes are normalized and raw detector dimensions remain private.

## 5.4 Common mask

The 180 s signal-rate image defines a source mask that is dilated to cover the local feature footprint. Linearity uses only pixels below 55,000 ADU in every exposure.

## 5.5 Response model

Pedestal-subtracted counts in the common mask are summed and fitted through the origin against exposure time. This is an exposure-response diagnostic, not a complete camera-linearity calibration.

## 5.6 Registration

Block-averaged log images are compared by phase correlation, with parabolic interpolation around the peak. The model fits translation only, not rotation, scale, or local distortion.

## 5.7 HDR estimator

At each pixel, the longest exposure below 60,000 ADU is converted to signal per second. Shorter exposures replace longer values near the adopted ceiling.

## 5.8 Candidates

Local maxima after mild Gaussian smoothing are ranked. The 500 displayed candidates are morphological detections and must not be interpreted as 500 identified atomic lines.

> **HDR rate estimator**
> $R(x,y) = [I_{t^*}(x,y) - B_{t^*}] / t^*$
> where $t^*$ is the longest exposure for which $I_{t^*}$ is below 60,000 ADU.

# 6 Quantitative exposure results

| Exposure | Background | Robust sigma | Bright pixels | >=60,000 ADU | Signal / 30 s |
|---|---|---|---|---|---|
| 30 s | 502 | 2.965 | 6,267 | 1 | 1.00000 |
| 60 s | 504 | 4.448 | 10,906 | 15 | 1.97961 |
| 120 s | 505 | 4.448 | 17,998 | 75 | 4.01202 |
| 180 s | 508 | 4.448 | 26,132 | 121 | 5.84104 |

The background pedestal changes by only 6 ADU across the sequence. Visible structure increases with exposure, but the near-ceiling population rises from 1 to 121 pixels. The integrated values remain close to the ideal exposure ratios 1:2:4:6.

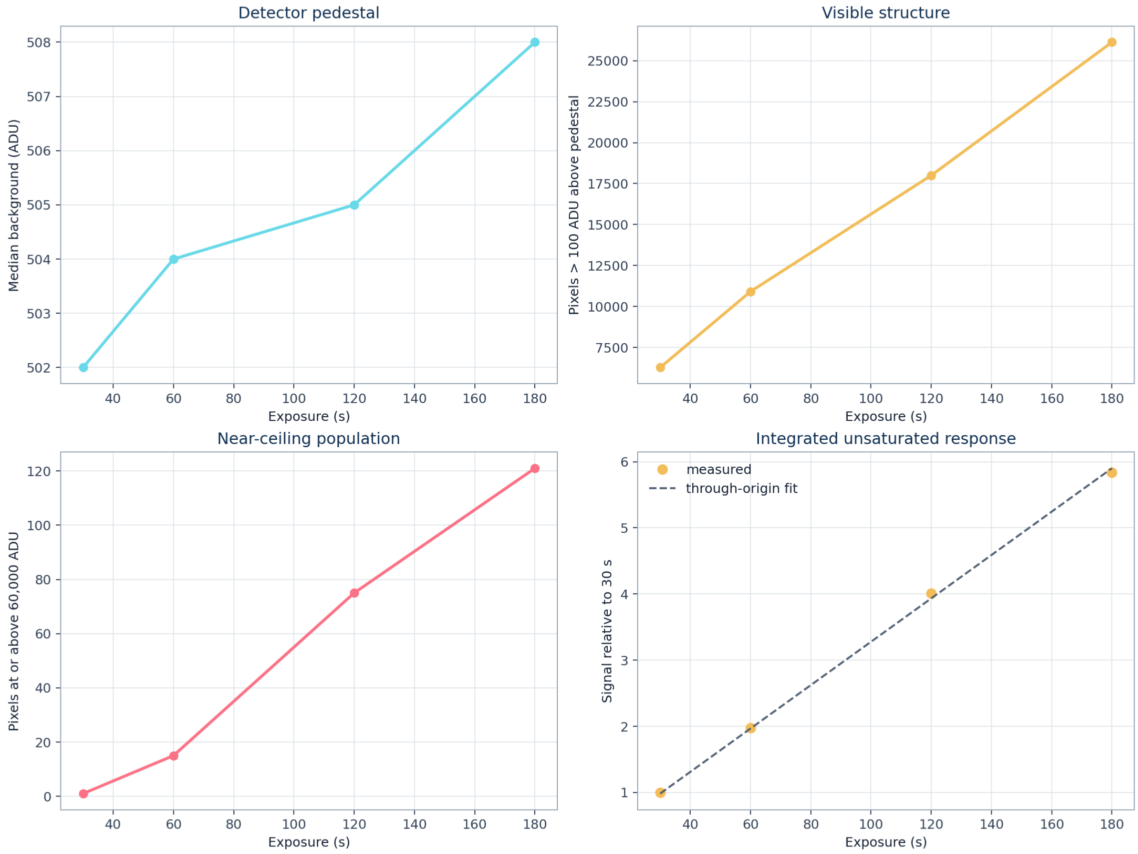


Figure 3. Detector pedestal, visible structure, near-ceiling population, and integrated unsaturated response. The common-mask response gives $R^2$ = 0.999276.

# 7 Registration and profile structure

Translation estimates relative to the 120 s frame are (-0.070, -0.012), (-0.065, -0.017), (0, 0), and (-0.045, -0.019) pixels in (y, x) for 30, 60, 120, and 180 s. The maximum magnitude is 0.071 pixel. These values show consistency under the algorithm, not a full uncertainty budget.

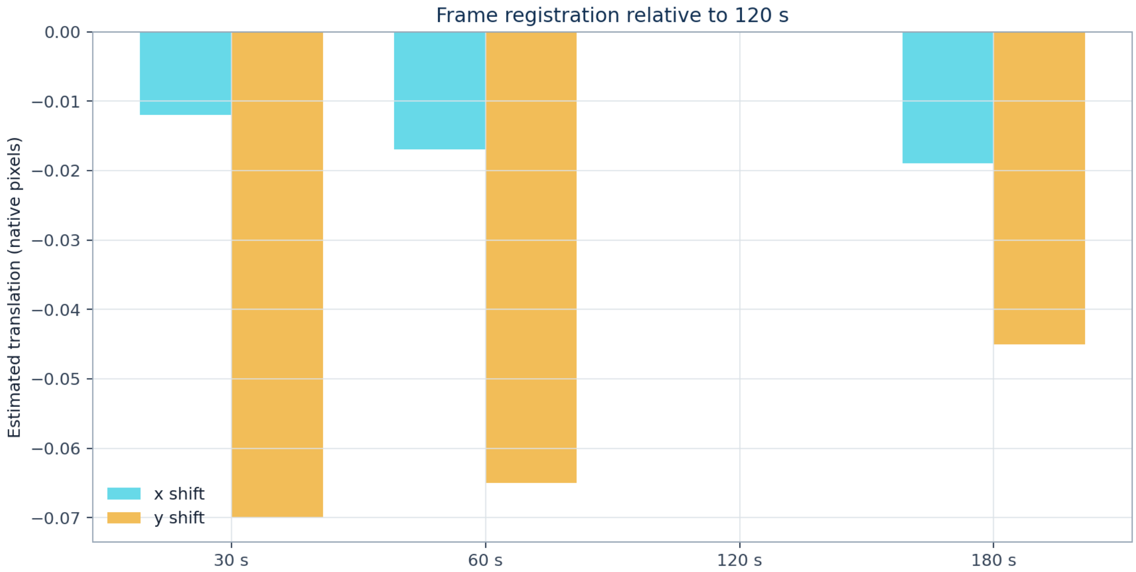


Figure 4. Phase-correlation shifts relative to 120 s after four-pixel block averaging.

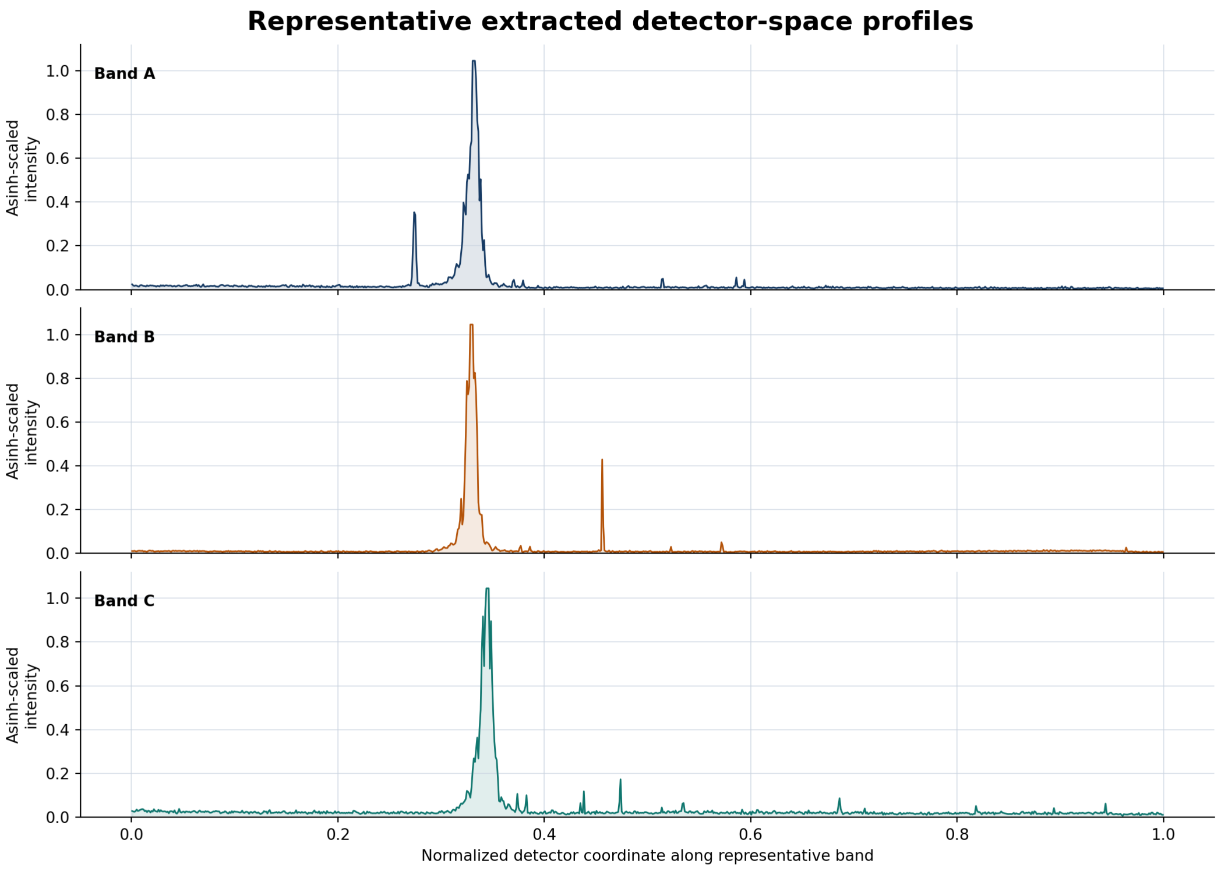


Figure 5. Representative detector-space profiles with asinh-scaled intensity. The horizontal coordinate is normalized detector position, not wavelength.

# 8 Interpretation and exposure recommendation

The 120 s exposure is sufficient for a single public example and is the recommended single frame for this sequence. It reveals much more faint structure than 30 or 60 s while retaining more bright-core headroom than 180 s. This result is specific to the acquisition conditions and must not be treated as a universal EXOhSPEC setting.

For HDR construction the 180 s frame supplies 66,279 valid source-mask pixels. The estimator falls back to 120 s for 49 pixels, 60 s for 57 pixels, and 30 s for 14 pixels. The small fallback population carries disproportionately important bright-core information.

## Recommended acquisition practice

- Retain 120 s as the main exposure for this configuration.
- Retain a 30 s exposure to protect the strongest feature cores.
- Retain 180 s where faint-feature visibility is important.
- Repeat each duration if temporal repeatability and random uncertainty are required.
- Do not increase beyond 180 s before reviewing near-ceiling behaviour and line-core linearity.
- Keep complete calibration metadata privately even when public derivatives omit it.

The present four frames are enough for this detector-domain report. Additional repeated observations would be needed for cosmic-ray rejection, lamp warm-up characterization, drift versus time, or a statistically defensible repeatability error.

# 9 Path to a wavelength-calibrated Th-Ar atlas

- 1. Determine the dispersion and cross-dispersion orientation.
- 2. Trace the centre and width of every usable echelle order.
- 3. Model local inter-order background and extract one-dimensional spectra with uncertainties.
- 4. Obtain an approximate order and wavelength seed from the optical model or prior calibration.
- 5. Centroid isolated, unsaturated features and estimate centroid uncertainties.
- 6. Match candidates to a versioned NIST Th-Ar reference list within physical tolerances.
- 7. Reject blends, unsuitable carrier-gas lines, saturated features, and statistical outliers.
- 8. Fit a two-dimensional wavelength model across pixel coordinate and order number.
- 9. Inspect signed residuals versus order, wavelength, detector position, intensity, and species.
- 10. Validate on withheld lines or an independent reference before publishing identifications.

A small global polynomial residual is not sufficient evidence of a correct solution. Local residual structure, blends, air-versus-vacuum convention, reference-list version, weights, and clipping rules must all be documented. Until this chain is complete, a wavelength axis in angstroms would be scientifically misleading.

# 10 Limitations

- Only one frame exists at each duration; fixed-exposure repeatability is unknown.
- The response fit assumes stable lamp output across the sequence.
- Registration models only global translation.
- The 60,000 ADU ceiling is an adopted diagnostic threshold, not an independently calibrated linearity limit.
- The public background treatment is simpler than a full local scattered-light model.
- Feature candidates are not atomic identifications.
- No claim is made about absolute flux, resolving power, line-spread function, or radial-velocity precision.

# 11 Applications to astronomical spectroscopy

| Application | Role of Th-Ar |
|---|---|
| **Absolute wavelength calibration** | Reference wavelengths convert detector coordinates into a physical wavelength scale. |
| **Instrument drift** | Repeated lamp frames monitor motion of the spectral format. |
| **Radial velocity** | Reliable wavelength coordinates are essential before small stellar Doppler shifts are interpreted. |
| **Extraction diagnostics** | The line-rich pattern tests trace, background, and order-extraction consistency. |
| **Resolution and line shape** | Isolated unsaturated features can constrain widths, asymmetry, and focus variation. |
| **Hybrid calibration** | Th-Ar absolute anchors may be paired with a Fabry-Perot etalon or laser-frequency comb. |

In exoplanet work the lamp does not detect a planet. It establishes the instrumental wavelength coordinate against which stellar motion or atmospheric structure can be measured. Calibration uncertainty propagates into the astrophysical result.

# 12 Conclusions

The supplied EXOhSPEC sequence is sufficient for a high-level scientific report. It demonstrates a dense emission-line field and quantitatively captures the exposure trade between faint-feature reach and bright-core headroom. Response is close to proportional over 30-180 s on the common unsaturated mask, and global registration is stable under the stated phase-correlation model.

Use 120 s when only one frame can be shown, but retain the full exposure ladder for HDR display and protection of bright cores. Do not publish atomic labels or wavelengths until trace extraction, line matching, the dispersion solution, and residual validation are complete.

ACKNOWLEDGEMENTS Supervision and contribution

This work was carried out by **Biswajit Jana** in the context of EXOhSPEC instrumentation research. The author acknowledges **Prof. Hugh Jones** for supervision of the EXOhSPEC research and **Prof. Bill Martin** for supervision of the optics and laboratory work at the University of Hertfordshire.

The public release was prepared to communicate the scientific analysis while respecting the privacy of the detector configuration, optical design implementation, laboratory layout, FITS headers, and raw data.

## REFERENCES Scientific and data sources

1. Lhospice, E. et al. (2019). *EXOhSPEC folded design optimization and performance estimation*. Proceedings of SPIE. Public manuscript.

2. University of Hertfordshire. EXOhSPEC project page.

3. Nave, G. et al. *Spectrum of Th-Ar Hollow Cathode Lamps*. NIST SRD 161, DOI 10.18434/T4S01V. NIST database.

4. Redman, S. L., Nave, G., and Sansonetti, C. J. (2014). The Spectrum of Thorium from 250 nm to 5500 nm. *ApJS* 211(1). DOI 10.1088/0067-0049/211/1/4.

5. Errmann, R. et al. (2020). HiFLEx: A Highly Flexible Package to Reduce Cross-dispersed Echelle Spectra. *PASP* 132, 064504. DOI 10.1088/1538-3873/ab8783.

6. Lovis, C. and Pepe, F. (2007). A new list of thorium and argon spectral lines in the visible. *Astronomy and Astrophysics*.

7. US Geological Survey. Thorium in the upper continental crust and soils.

8. NOAA National Weather Service. Composition of the atmosphere.

9. Commission on Isotopic Abundances and Atomic Weights. Thorium isotopic composition.

10. NIST. Atomic weights and isotopic compositions for argon.

11. ESO. ESPRESSO wavelength-calibration quality control.

Reference line databases are cited as the intended provenance for a future wavelength solution. Their existence does not by itself identify peaks in the present detector images.